

Detailed Explanation of KDE and PDF Estimation for Inter-Pellet Interval Analysis

Introduction

Kernel Density Estimation (KDE) and Probability Density Function (PDF) estimation are essential for analysing inter-pellet intervals (IPIs) derived from FED3 data. These techniques allow for visualising and understanding the temporal patterns of feeding behaviour in mice, providing insights into their dietary habits and experimental conditions.

This document elaborates on each component of KDE and PDF estimation, explaining its relevance and providing concrete examples of how it is applied to IPIs.

Elements of KDE and PDF Estimation for IPI Analysis

1. Kernel Density Estimation (KDE)

KDE is a non-parametric approach to estimating the probability density function of a random variable, in this case, the inter-pellet intervals (IPIs). By applying KDE, we obtain a smooth curve that approximates the underlying distribution of IPIs without assuming any specific parametric model.

Mathematical Definition:

$$\hat{f}(x) = \frac{1}{nh} \sum_{i=1}^n K\left(\frac{x - x_i}{h}\right),$$

where:

- n is the number of IPI data points.
- x_i is a specific IPI value from the dataset.
- h is the bandwidth controlling smoothness.
- K is the kernel function (e.g., Gaussian).

Example: Suppose the IPIs are {1.2, 1.5, 2.3, 1.8, 3.0} seconds, and we want to estimate the density at $x = 2.0$. For a Gaussian kernel:

$$K(u) = \frac{1}{\sqrt{2\pi}} e^{-\frac{u^2}{2}},$$

we calculate $K\left(\frac{2.0-x_i}{h}\right)$ for each x_i , sum them, and scale by $1/(nh)$.

How it helps:

- Highlights the most common IPI durations by showing peaks in the density curve.
- Smoothens raw data to make trends more apparent, especially in noisy datasets.
- Allows comparison of density shapes across experimental groups (e.g., diet phases).

2. Bandwidth Selection

The bandwidth parameter determines the degree of smoothing applied to the KDE. A small bandwidth captures finer details but may overfit, while a large bandwidth smooths out noise but risks losing important features.

Example: Using the same IPIs as above, with $h = 0.5$, the density estimate at $x = 2.0$ would be more sensitive to nearby IPIs. Increasing h to 1.0 would result in a smoother curve but less detail.

How it helps:

- Ensures that the density curve accurately represents the distribution of IPIs.
- Aids in identifying specific feeding patterns, such as frequent short intervals or occasional long gaps.

3. Kernel Function

The kernel function determines how each data point contributes to the overall density. A commonly used kernel is the Gaussian function, which weights data points based on their proximity to the evaluation point.

Example: For $x = 2.0$ and $x_i = 1.8$, the kernel value for a Gaussian function is:

$$K\left(\frac{2.0 - 1.8}{h}\right) = \frac{1}{\sqrt{2\pi}} e^{-\frac{(0.2/h)^2}{2}}.$$

This contribution decreases as the distance $|x - x_i|$ increases.

How it helps:

- Provides a smooth and continuous estimate of the IPI distribution.
- Ensures that distant data points have less influence on the density estimation, reflecting localised patterns.

4. Probability Density Function (PDF)

The PDF is the output of KDE, representing the likelihood of observing a specific IPI value. For feeding behaviour analysis, it provides a continuous representation of how often particular intervals occur.

Example: After applying KDE to the given IPIs, the PDF curve shows peaks at common intervals like 1.5 and 2.3 seconds, with smaller values elsewhere.

How it helps:

- Visualises the most frequent feeding intervals.
- Helps to identify multimodal distributions, indicating different behavioural states.
- Allows comparison of distributions across groups (e.g., protein-restricted vs non-restricted diets).

5. Jittered Data Points

Jittering involves adding small random offsets to data points when plotting them. This technique is especially useful when individual data points overlap, making it difficult to discern their density.

Example: Plotting the IPIs with jitter along the x -axis separates overlapping points like 1.2 and 1.3, making the density more apparent.

How it helps:

- Reveals the distribution of raw data points, complementing the KDE curve.
- Highlights areas of high data concentration without obscuring the KDE.

6. Heatmaps for Data Point Density

Heatmaps use colour gradients to represent the density of data points on a plot. In the context of IPI analysis, they provide an additional visual layer showing where the data clusters.

Example: A heatmap overlaying the KDE highlights regions like $x = 1.5$ and $x = 2.3$ where data density is highest, using warm colours like red for high density and blue for low density.

How it helps:

- Enhances the visualisation of data density, especially in areas with overlapping points.
- Highlights feeding patterns, such as frequent short IPIs or rare long IPIs.

7. Group Comparisons

Comparing KDE curves and PDFs between groups (e.g., diet phases or orders) highlights differences in feeding behaviour. Overlaying curves allows direct visualisation of variations in IPI distributions.

Example: Comparing KDEs of protein-restricted and non-restricted groups shows a shift in peaks, with one group having shorter IPIs.

How it helps:

- Identifies shifts in feeding patterns due to experimental conditions.
- Quantifies differences between groups using statistical metrics (e.g., Kolmogorov-Smirnov test).
- Facilitates interpretation of the impact of diet or order on feeding intervals.

Application to FED3 Data

1. **Data Preprocessing:** Extract IPI data from FED3, ensuring proper grouping by diet phases and orders.
2. **KDE Estimation:** Apply KDE to visualise the distribution of IPIs for each group.
3. **Overlay Visual Elements:** Add jittered data points and heatmaps to enhance interpretation.
4. **Group Analysis:** Compare KDE curves between groups to identify behavioural differences.
5. **Statistical Validation:** Use statistical tests to confirm observed differences.

Conclusion

Each element of KDE and PDF estimation contributes to a comprehensive understanding of inter-pellet interval distributions. By integrating KDE, bandwidth optimisation, jittered points, and heatmaps, researchers can uncover subtle patterns in feeding behaviour, providing valuable insights into the effects of diet and experimental conditions on mice.